MIL-P-46431C (PA)
20 August 1976
SUPERSEDING
MIL-P-46431B (MU)
9 August 1965

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M86
PARTS FOR

This specification is approved for use by Picatinny Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 This specification covers the parts for one type of electric primer designated as Primer, Electric, M86.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, packing and Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1390

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DRAWINGS

ARMY

8847425 - Body, Metal Parts Assembly
9313896 - Body, Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts assembly.-The metal parts assemblies shall comply with all requirements specified on Drawing (dwg.) 8847425 or 9313896 and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.5.1 (see 6.3).

3.4 Pull test of brazed joint.-The brazed joint securing the head to the primer body shall withstand a pull of 5500 pounds min. without evidence of separation of the head from the primer body when tested as specified in 4.5.2.

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.6 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 25 each individual components, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirement of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes, and tests. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

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4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46431C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Head				7548316
					NEXT HIGHER ASSEMBLY 8847425
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:	None defined				
Minor:					
201	Total length		.65%	3.2	Gage
202	Depth of body counterbore		.65%	3.2	Gage
203	Concentricity of body counterbore with external head thread		.65%	3.2	Gage
204	Evidence of poor wrokmanship		.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46431C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Head alternate		9313898
			NEXT HIGHER ASSEMBLY 9313896

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of external head thread, min.		.40%	3.2	Gage
102	Major diameter of external head thread, min.		.40%	3.2	Gage
103	Pitch diameter of internal head thread, max.		.40%	3.2	Gage
104	Minor diameter of internal head thread, max.		.40%	3.2	Gage
105	Depth of internal head thread counterbore		.40%	3.2	Gage
106	Protective coating missing or damaged		.40%	3.2	Gage
107	Concentricity of internal head thread with external head thread		.40%	3.2	Gage
108	Pitch diameter of thread for body max.		.40%	3.2	Gage
109	Minor diameter of thread for body max.		.40%	3.2	Gage
110	Concentricity of thread for body with external head thread		.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9313898
4.4.2.2	(Continuation)				NEXT HIGHER ASSEMBLY 9313896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor:					
201	Total length		.65%	3.2	Gage
202	Length of external head thread, including thread undercut, min.		.65%	3.2	Gage
203	Width of external head thread undercut, max.		.65%	3.2	Gage
204	Depth of internal head thread, min.		.65%	3.2	Gage
205	Diameter of external head thread undercut		.65%	3.2	Gage
206	Diameter of internal head thread counterbore		.65%	3.2	Gage
207	Width of wrench slots, min.		.65%	3.2	Gage
208	Depth of wrench slots		.65%	3.2	Gage
209	Inside diameter of head		.65%	3.2	Gage
210	Outside diameter of head		.65%	3.2	Gage
211	Thickness of flange		.65%	3.2	Gage
212	Diameter of flange, max.		.65%	3.2	Gage
213	Pitch diameter of external head thread, max.		.65%	3.2	Gage
214	Major diameter of external head thread, max.		.65%	3.2	Gage
215	Concentricity of internal head thread counterbore with internal head thread		.65%	3.2	Gage
216	Concentricity of flange with external head thread		.65%	3.2	Gage
217	Pitch diameter of internal head thread, min.		.65%	3.2	Gage
218	Marking, missing, incorrect or unidentifiable		.65%	3.2	Visual
219	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46431C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Body		8847426
			NEXT HIGHER ASSEMBLY 8847425

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1	One or more flash holes missing		100%	3.2	Gage
Major:					
101	Plug gage fails to pass through body		.40%	3.2	Gage
102	Location of first row of holes		.40%	3.2	Gage
103	Location of second row of holes		.40%	3.2	Gage
104	Wall thickness		.40%	3.2	Gage
105	Diameter of flash holes		.40%	3.2	Gage
106	Hydrostatic Test	4.4.3.1		3.3	4.5.1
Minor:					
201	Total length		.65%	3.2	Gage
202	Outside diameter, max.		.65%	3.2	Gage
203	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46431C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9313897
4.4.2.4	Body Alternate				NEXT HIGHER ASSEMBLY 9313896
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1	One or more flash holes missing		100%	3.2	Gage
2	Crack or split		100%	3.2	Visual
Major:					
101	Pitch diameter of thread, min.		.40%	3.2	Gage
102	Major diameter of thread, min.		.40%	3.2	Gage
103	Outside diameter, max.		.40%	3.2	Gage
104	Wall thickness, min.		.40%	3.2	Gage
105	Diameter of holes		.40%	3.2	Gage
106	Location of holes		.40%	3.2	Gage
107	Plug gage fails to pass through body		.40%	3.2	Gage
108	Hydrostatic Test	4.4.3.1		3.3	4.5.1
Minor:					
201	Pitch diameter of thread, max.		.65%	3.2	Gage
202	Wall thickness, max.		.65%	3.2	Gage
203	Length of thread, min.		.65%	3.2	Gage
204	Protective finish with bare spot		.65%	3.2	Visual
205	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Body, metal parts assembly				8847425
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1	Crack or split (prior to plating).		100%	3.2	Visual
Major:					
101	Pitch diameter of external head thread, min.		.40%	3.2	Gage
102	Major diameter of external head thread, min.		.40%	3.2	Gage
103	Pitch diameter of internal head thread, max.		.40%	3.2	Gage
104	Minor diameter of internal head thread, max.		.40%	3.2	Gage
105	Depth of internal head thread counterbore		.40%	3.2	Gage
106	Braze not full 360 degrees		.40%	3.2	Visual
107	Protective coating missing or damaged		.40%	3.2	Visual
108	Concentricity of internal head thread with external head thread		.40%	3.2	Gage
109	Braze joint	4.4.3.2		3.4	4.5.2
Minor:					
201	Length from face of flange to end of external head thread, min.		.65%	3.2	Gage
202	Width of external head thread undercut, max.		.65%	3.2	Gage
203	Depth of internal head thread min.		.65%	3.2	Gage
204	Diameter of external head thread undercut		.65%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46431C (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5	(Continuation)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
205	Diameter of internal head thread counterbore		.65%	3.2	Gage
206	Width of wrench slots, min.		.65%	3.2	Gage
207	Depth of wrench slots		.65%	3.2	Gage
208	Inside diameter of head		.65%	3.2	Gage
209	Outside diameter of head		.65%	3.2	Gage
210	Thickness of flange		.65%	3.2	Gage
211	Diameter of flange, max.		.65%	3.2	Gage
212	Pitch diameter of external head thread, max.		.65%	3.2	Gage
213	Major diameter of external head thread, max.		.65%	3.2	Gage
214	Total length, max.		.65%	3.2	Gage
215	Concentricity of internal head thread counterbore with internal head thread		.65%	3.2	Gage
216	Concentricity of flange with external head thread		.65%	3.2	Gage
217	Pitch diameter of internal head thread, min.		.65%	3.2	Gage
218	Solder on interior surfaces		.65%	3.2	Visual
219	Marking, missing, misleading or unidentifiable		.65%	3.2	Visual
220	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Metal parts assembly alternate				9313896
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101	Total length		.40%	3.2	Gage
102	Torque disassembly		.40%	3.2	Test

NOTES:

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4.4.3 Testing

4.4.3.1 Hydrostatic test (see 3.3) - Critical defect - This test shall be conducted 100 percent in accordance with the procedure specified in 4.5.1. If any primer tubing fails to comply with the requirement specified, it shall be classed defective and no primer bodies shall be fabricated therefrom.

4.4.3.2 Pull test of brazed joint (see 3.4) - Critical defect.

4.4.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 150 body assemblies shall be randomly selected from each lot for this test. If any body fails to comply with the applicable requirement the lot shall be rejected. The test shall be as specified in 4.5.2 using equipment in accordance with 4.4.4.

4.4.3.2.2 After three consecutive lots have met the criteria specified in 4.4.3.2.1, 50 body assemblies shall be randomly selected from each lot. If any body fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2 using equipment in accordance with 4.4.4.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in Paragraph Reference/Inspection Method Column in tables starting with paragraph 4.4.2.1 and 4.3. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6. of MIL-A-48078.

4.5 Test methods and procedures

4.5.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

P=1.8 St divided by D

where:

P = tests pressure in pounds per square inch (psi)
S = minimum specified yield strength, psi
t = specified minimum wall thickness in inches
D = specified maximum outside diameter in inches

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4.5.2 Pull test of brazed joint.-The body assembly shall be assembled to the applicable test fixture specified in 4.4.4. The specified load shall be applied in an axial direction. The rate of travel of the testing machine shall be 0.100 to 0.125 inch per minute. The sample assemblies subjected to this test shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-The parts shall be packaged, packed and marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of design for approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Hydrostatic Test.-If the primer tubing procured for the bodies manufactured in accordance with this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1390-A144

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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